

Q1 short-answer section was removed in publishable version.

(60 points)

Question 1. Short Answer Questions

Fill in the box with the answer.

- *Removed in publishable version* -

Question 2. Coding 1

Q2 addFirst(T item): count existing equal items; if count >= 1 throw IllegalArgumentException; otherwise super.addFirst(item).

(20 points)

Complete the overridden method void addFirst(T item).

It is a method of the class DelimitedARDeque, a subclass of an iterable ARDeque.

It adds an item to the front of the object of DelimitedARDeque,

but it will throw an IllegalArgumentException if the item already occurs in it more than once.

Test case:

```
DelimitedARDeque<String> dd = new DelimitedARDeque<>();
dd.addFirst("a");
dd.addFirst("b");
try {
    dd.addFirst("a");
    dd.printDeque();
} catch (IllegalArgumentException e) {
    System.out.println("AddFirst failed");
}
```

→ AddFirst failed

Q3 equals(Object): same compatible deque
+ same size + at most 1 different item at
same position.

(20 points)

Question 3. Coding 2

Complete the method `boolean equals(Object that)` of an iterable `ARDeque` that overrides the one in `Object`.

To be the same, they must be of the same size, and must share at most 1 different items at the same position.

Suppose `[]` is an `ARDeque<Integer>`. Then:

`[1, 2, 3].equals([1, 2, 5])` → `true`

`[1, 3, 2].equals([1, 2, 3])` → `false`

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